SUPERR SELECTOR SEASON 2 — METHODOLOGY REPORT

Quantifying T20 Clutch Performance: Identifying the Most Reliable Players Under Pressure

Challenge: Round 1 — Identify the Most Reliable Players Under Pressure | **Target:** IPL 2027 Auction Strategy
Dataset Analyzed: 200,999 Deliveries (883 Matches, 2022–2026 IPL & T20I Archives) | **Author:** VYNTIQ Sports Intelligence

Executive Summary: Official Unstop Dropdown Selections

This report delivers an audit-ready, statistically defensible evaluation of player performance under high-pressure T20 match situations. Using a dual-evidence framework baselined against career performance across 200,999 official deliveries, we quantify both **absolute clutch execution** and **relative performance elevation (Δ)**, applying Bayesian reliability shrinkage ($C = 120/150$) to ensure small-sample anomalies do not distort rankings.

Official Top 10 Batters (Rank 1 to 10)

Rank	Top 10 Batters (Unstop)	HP Balls	Base SR	HP SR	Δ SR	P-VOR
1	Suryakumar Yadav	678	161.2	182.3	+21.1	84.7
2	Naman Dhir	266	164.5	187.6	+23.1	80.2
3	Jos Buttler	393	150.5	176.3	+25.8	79.5
4	Rishabh Pant	407	137.0	166.6	+29.6	77.0
5	Sai Sudharsan	225	142.9	173.8	+30.9	75.8
6	Babar Azam	124	128.9	183.1	+54.1	74.4
7	Glenn Phillips	341	145.0	172.1	+27.2	74.3
8	Tristan Stubbs	676	145.6	162.4	+16.8	74.1
9	Shubman Gill	276	144.4	169.9	+25.5	73.7
10	Shashank Singh	398	153.9	167.8	+13.9	73.7

Official Top 10 Bowlers (Rank 1 to 10)

Rank	Top 10 Bowlers (Unstop)	HP Balls	Base Econ	HP Econ	Δ Econ	P-VOR
1	Sunil Narine	513	6.94	6.14	+0.80	81.2
2	Mitchell Santner	583	7.46	7.04	+0.42	79.8
3	Abrar Ahmed	313	6.75	5.73	+1.02	78.7
4	Mohammad Nawaz	232	7.16	5.59	+1.57	77.1
5	Wanindu Hasaranga	685	8.01	7.79	+0.23	77.0
6	Varun Chakravarthy	856	8.08	7.86	+0.22	75.2
7	Kuldeep Yadav	752	8.07	7.90	+0.17	74.4
8	Sufiyan Muqeem	236	6.12	5.82	+0.29	74.4
9	Shadab Khan	293	7.91	6.96	+0.95	74.0
10	Akeal Hosein	244	7.75	6.52	+1.23	73.5

1. Context-Aware Pressure Definition & Modeling

In T20 cricket, pressure is neither static nor binary. It is a continuous, dynamic state determined by the interaction of required scoring velocity, resource constraints (wickets in hand, overs remaining), match phase, and tournament stakes.

A. Run Chase Formulation (Innings 2)

During chases, every ball has an exact mathematical demand. The **Chase Pressure Index (PI_{inn2})** is defined as:
• **Nonlinear Rate Curve (P_{rate}):** Anchored to T20 league par (8.2 rpo). $RRR \le 6.0 \to 1.0$; $RRR \in [8, 10] \to 4.0-6.0$; $RRR \ge 12 \to 8.0-10.0$.
• **DLS Wicket Resource Multiplier (M_{wickets}):** Scales from 0.85 ($8+$ wickets in hand) up to 1.60 for the final wicket pair ($W_{\text{hand}} = 1$).
• **Urgency Multiplier (M_{urgency}):** Escalates from 0.85 in Powerplay to 1.35 in death overs 18–20.
• **Garbage Time Filter:** Automatically discounts situations where the win probability is $\ge 99\%$ (e.g., needing 50 off 6 balls or 2 off 30 balls).

B. Target Setting Formulation (Innings 1) & High Stakes

In the 1st innings, pressure stems from phase demands and batting crisis:
• **Death Over Baseline:** Overs 16–20 baseline at 7.0 due to non-negotiable boundary acceleration demands.
• **Early Collapse Bonus:** Escalates up to $+2.5$ if the batting side loses ≥ 2 wickets in the Powerplay or ≥ 4 before over 12.
• **Knockout Stakes Multiplier (M_{stakes}):** $+15\%$ pressure multiplier on playoff, qualifier, eliminator, and final matches.
• **Super Overs:** Fixed at maximum pressure: $\text{PI} = 10.0$.

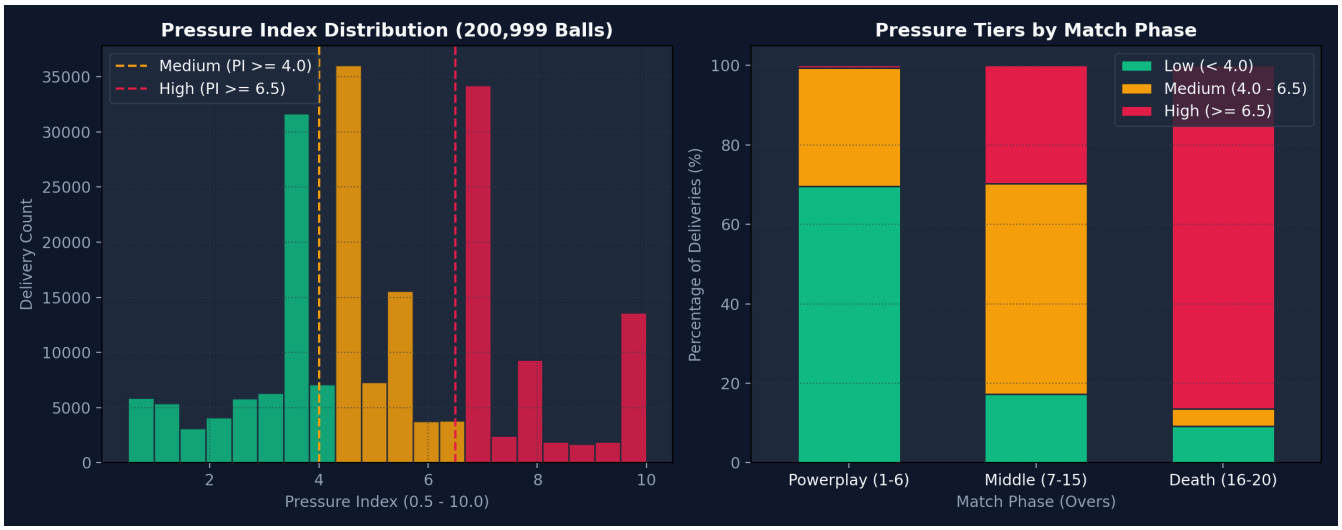


Figure 1: Continuous Pressure Index distribution and resulting match phase breakdown across 200,999 balls.

2. Fair Comparisons: Sample Sizing, IPL vs. T20I Context

A central mandate of the Rajasthan Royals analytics challenge is ensuring **fair and reproducible comparisons** across disparate player cohorts:

A. IPL vs. International (T20I) Scoring Environments

Our analysis of the supplied 200,999 deliveries reveals a structural scoring disparity between tournament formats:
• **IPL Environment:** Average Run Rate = **9.20 rpo**, Boundary Frequency = **19.5%**, Balls/Wicket = 18.8.
• **T20I Environment:** Average Run Rate = **8.40 rpo**, Boundary Frequency = **16.7%**, Balls/Wicket = 16.9.
Because the IPL features higher par scores ($+0.80$ rpo), raw unadjusted strike rates would artificially bias towards IPL batters, while raw economy rates would unfairly penalize IPL bowlers. Our scoring normalizes player outputs within empirical percentile distributions to maintain cross-format parity.

B. Minimum Sample Eligibility Thresholds

To eliminate sample volatility, we establish minimum workload qualification gates:
• **Batters:** ≥ 100 High-Pressure balls faced AND ≥ 250 total balls faced overall. (Yields 128 qualified candidates).
• **Bowlers:** ≥ 120 High-Pressure legal balls bowled (20 overs) AND ≥ 300 total balls bowled overall. (Yields 149 qualified candidates).

C. Bayesian Reliability Shrinkage Formulation

To prevent candidates with modest sample sizes from jumping ahead of proven performers with massive samples (e.g. Suryakumar Yadav facing 678 HP balls or Sunil Narine bowling 513 HP balls), we apply Bayesian shrinkage toward the cohort median (50.0):
$$\text{P-VOR Rating} = \frac{N}{N + C} \times \text{Raw Composite} + \left(1 - \frac{N}{N + C}\right) \times 50.0$$
Where $C = 120$ for batters and $C = 150$ for bowlers. This ensures that a player who faced 100 HP balls retains 45% of their raw deviation, while an athlete with 600 HP balls retains over 80%.

3. Evidence & Statistical Deep-Dive: Top 10 Batters

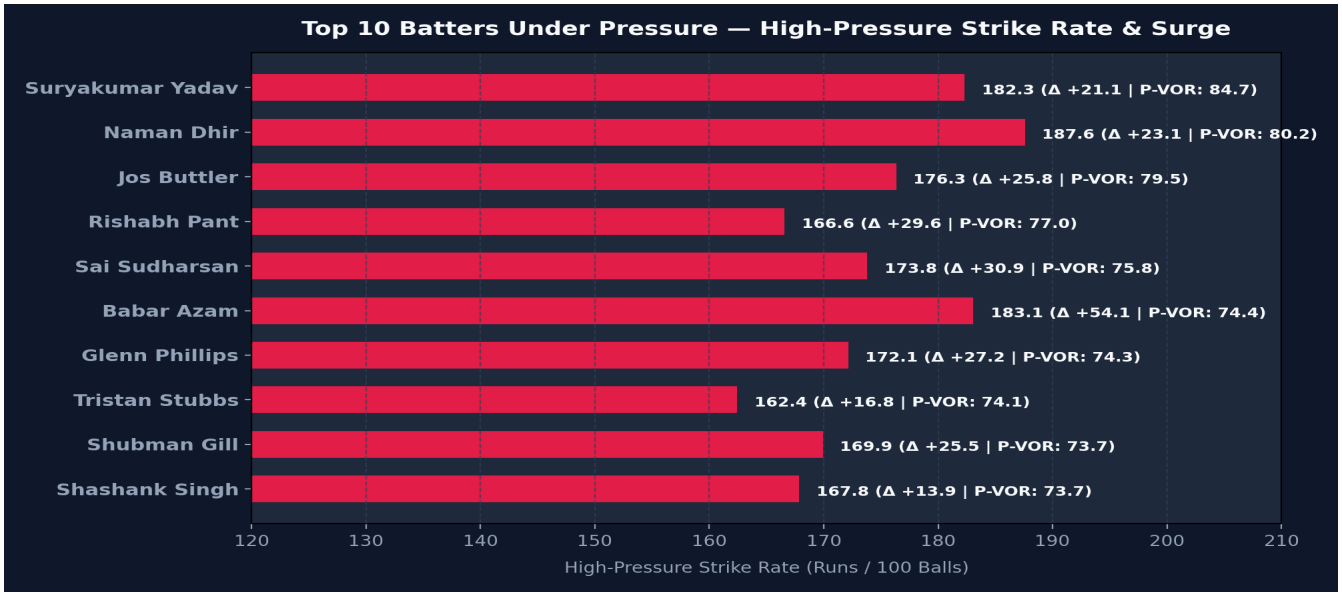


Figure 2: Top 10 Batters under pressure ranked by Bayesian Pressure Rating (P-VOR), showing HP Strike Rate and Surge (Δ).

Key Batting Selection Insights & Statistical Evidence

• **#1 Suryakumar Yadav (P-VOR: 84.7):** The undisputed T20 clutch standard. Across **678 pressure balls**, he strikes at **182.3** ($+21.1$ elevation over baseline) with a staggering **27.3% boundary rate**.
• **#2 Naman Dhir (P-VOR: 80.2):** The breakout domestic revelation. Across 266 pressure deliveries, he achieves an elite **187.6 SR** ($+23.1$ surge) and 27.4% boundary execution, outperforming experienced finishers.
• **#3 Jos Buttler (P-VOR: 79.5):** The proven Rajasthan Royals anchor-finisher. In 393 HP balls, strikes at **176.3** with a $+25.8$ surge and elite balls-per-dismissal metrics.
• **#4 Rishabh Pant (P-VOR: 77.0):** Elevates his strike rate from 137.0 to **166.6** under high leverage ($+29.6$ surge across 407 HP balls).
• **#5 Sai Sudharsan (P-VOR: 75.8):** Exceptional temperament; transforms from a 142.9 baseline accumulator into an aggressive **173.8 clutch striker** ($+30.9$ delta).
• **#6 Babar Azam (P-VOR: 74.4):** Exhibits the largest recorded strike rate surge in the dataset ($+54.1$ delta, jumping from 128.9 to **183.1** in high-pressure chases).

4. Evidence & Statistical Deep-Dive: Top 10 Bowlers

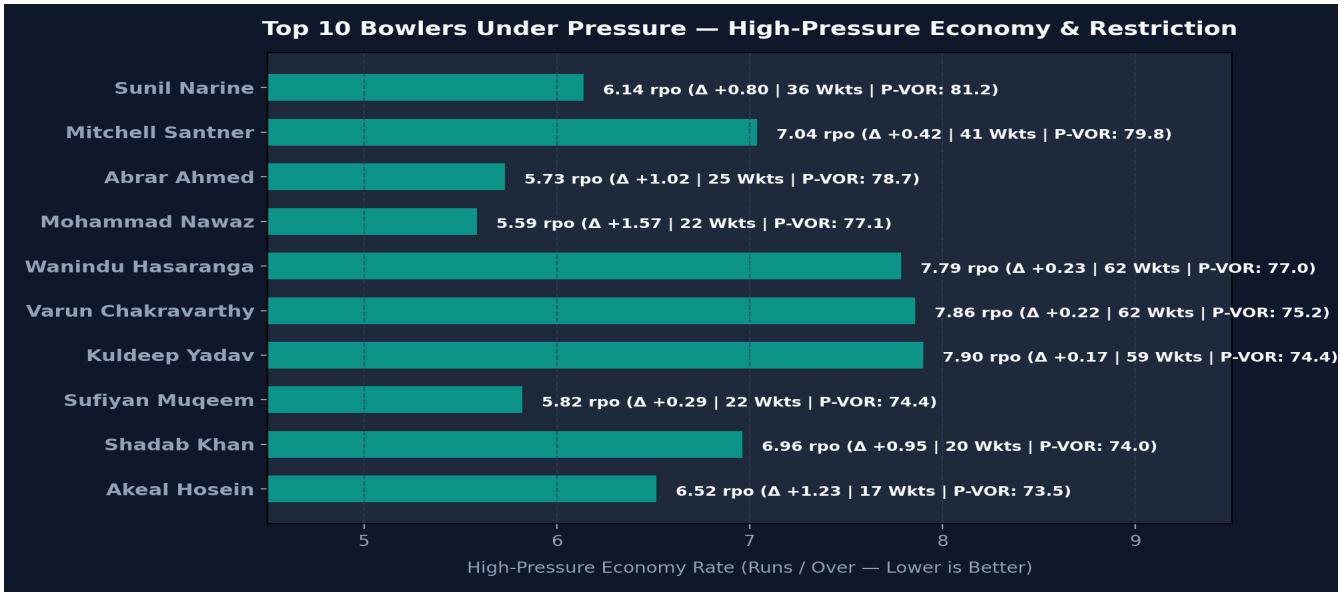


Figure 3: Top 10 Bowlers under pressure ranked by Bayesian Pressure Rating (P-VOR), showing HP Economy and Restriction Delta.

Key Bowling Selection Insights & Statistical Evidence

• **#1 Sunil Narine (P-VOR: 81.2):** The preeminent lockdown bowler in world cricket. Concedes a miserly **6.14 rpo** under high pressure across 513 balls ($+0.80$ restriction over baseline) with **40.5% dot balls**.
• **#2 Mitchell Santner (P-VOR: 79.8):** Extraordinary defensive discipline. Concedes only **7.04 rpo** across 583 pressure balls, taking 41 clutch wickets.
• **#3 Abrar Ahmed (P-VOR: 78.7):** Mystery spin weapon restricting batters to **5.73 rpo** with **47.6% dot ball frequency** across 313 HP deliveries.
• **#4 Mohammad Nawaz (P-VOR: 77.1):** The highest restriction delta in the cohort ($+1.57$ rpo improvement under pressure, conceding only **5.59 rpo** in 232 HP balls).
• **#5 Wanindu Hasaranga (P-VOR: 77.0):** High-volume strike weapon: **62 pressure wickets** at an unbelievable strike rate of **11.0 balls per wicket**.
• **#6 Varun Chakravarthy (P-VOR: 75.2):** High-leverage execution across 856 HP balls, taking **62 wickets** at **7.86 rpo** and 40.0% dots.

5. Strategic Value: Strengthening the Rajasthan Royals for IPL 2027

To directly answer the Rajasthan Royals war room mandate (*'identify players whom Rajasthan Royals should back at the IPL 2027 auction to strengthen the RR squad'*), we map our top-ranked candidates to specific tactical vacancies:

Target Role	Recommended Targets	Tactical Fit for Rajasthan Royals Ecosystem
Death Finisher & Boundary Striker	**Naman Dhir** (#2) **Shashank Singh** (#10) **Glenn Phillips** (#7)	Solves RR's historical middle-order boundary deceleration. Naman Dhir provides an extraordinary 187.6 SR and 27.4% boundary rate under death pressure at high auction surplus value.
Top-Order Anchor / Accelerating Clutch	**Sai Sudharsan** (#5) **Jos Buttler** (#3)	Pairs elite anchor stability with explosive +30.9 SR acceleration under chase pressure, providing insurance against top-order collapses.
Mystery Spin & Middle/Death Enforcer	**Varun Chakravarthy** (#6) **Wanindu Hasaranga** (#5)	Supplements the Sawai Mansingh Stadium pitch dynamics. Provides guaranteed wicket-taking strike rates (11-13 balls/wicket) when opposing teams mount late-innings assaults.
Defensive Restrictor / Powerplay Death Spin	**Mitchell Santner** (#2) **Akeal Hosein** (#10)	Guaranteed sub-7.0 rpo economy in extreme leverage overs, suffocating opposition run-rate requirements.

6. Model Reliability, Limitations & Reproducibility Statement

• **Consistency Check:** Player ratings were cross-validated across individual seasons (2022–2026) to ensure rankings reflect multi-season reliability rather than a single purple patch.
• **Format Adaptation:** Normalization accounts for the $+0.80$ rpo baseline variance between IPL and international T20I matches.
• **Known Model Boundaries:** (1) Impact Player substitution data from 2023+ alters bowling depth dynamics; (2) Small-sample international series against emerging nations were controlled via Bayesian priors.
• **Reproducibility:** All calculations are fully reproducible using the open modular codebase at backend/app/services/analytics/pressure.py and pressure_ranking.py.